

DIBAKAR ROY

[Portfolio](#)[dibakar0804](#)[roy056](#)[Google Scholar](#)

Panchagarh, Rangpur, Bangladesh

EDUCATION

• Rajshahi University of Engineering & Technology (RUET)

BSc. in Electrical & Computer Engineering

◦ CGPA: 3.39 / 4.00 | (Last 60 Credits: 3.99 / 4.00)

March 2022 - July, 2026

Rajshahi-6204, Bangladesh

RESEARCH INTERESTS

- Brain-Computer Interfaces (BCI)
- Intracortical Speech Decoding
- EEG Signal Processing
- Cross-Session Neural Adaptation
- Neural Engineering
- Neural Signal Processing
- Biomedical Machine Learning
- Adaptive Neural Decoding

RESEARCH EXPERIENCE

• Rapid Parameter-Efficient Calibration for Cross-Session Intracortical Speech Decoding

2025 – Present

Research Advisors: Fariya Tabassum and Rana Mozumder

- Developed a cross-session intracortical speech decoding framework on the public T12 and T15 datasets using strict chronological training, validation, and held-out future-session evaluation.
- Designed a parameter-efficient input-space calibration strategy that updates only 65,792 parameters ($<0.05\%$ of the complete decoder) while keeping the high-capacity BiGRU encoder and CTC classifier frozen.
- Reduced macro phoneme error rate on T12 by 11.37% and 15.51% using 40 and 80 calibration trials, respectively, with improvement across all six held-out sessions; independent T15 evaluation achieved up to 10.05% relative improvement.

• Baseline-Aware EEG Workload Classification with Reduced Electrodes

2026

- Developed a baseline-aware, leakage-safe framework for subject-independent workload classification on EEGMAT and STEW using nested leave-one-subject-out validation.
- Evaluated reduced and shared electrode montages, handcrafted and Riemannian features, and cross-dataset transfer, achieving balanced accuracies of 0.8199 on EEGMAT and 0.8519 on STEW.

• Machine Learning for RIS-Assisted Wireless Systems

2025

- Contributed primarily to the ML/DL components of two published IEEE studies on RIS-assisted ISAC and secure hybrid beamforming, including DRL-based joint beam/RIS optimization, DNN-assisted precoding, model training, simulation, and performance analysis.

PUBLICATIONS

- D. Roy, F. Tabassum, and R. Mozumder, "Rapid Parameter-Efficient Calibration for Cross-Session Intracortical Speech Decoding Across Two Public Datasets," *Scientific Reports*, Nature Portfolio. **Submitted**.

- D. Roy and F. Tabassum, "Fast Label-Free Cross-Session Calibration for Intra-Cortical Speech Decoding," 2026 [🔗](#)
• *IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, Chittagong, Bangladesh, 2026, pp. 1–6.

- D. Roy, D. Dhar, and R. Mozumder, "Baseline-Aware Practical Framework for EEG Workload Classification with Reduced Electrodes Across Multiple Datasets," *International Conference on Electrical, Computer and Communication Technologies (ECCCT)*, 2026.

Accepted and presented; camera-ready manuscript submitted in Taylor & Francis book-chapter format for publication.

- M. R. Maksud, D. Roy, P. R. Dipa, and M. A. H. Pramanik, "Hardware-Aware RIS-Assisted ISAC with Quantized Feedback: A Robust Deep Reinforcement Learning Framework," 2025 *28th International Conference on Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, 2025, pp. 5590–5595. [🔗](#)

- M. R. Maksud, D. Roy, M. M. Hussien, M. K. Ghosh, and M. F. Ahamed, "Learning-Augmented RIS-Aided Hybrid Beamforming for Secure and Low-Latency Transmission," 2025 *7th International Conference on Electrical Information and Communication Technology (EICT)*, Khulna, Bangladesh, 2025. [🔗](#)

ACADEMIC PROJECTS

• Event Management Website (Software Project II)

2024, RUET

- Built a centralized departmental event management system with admin/user dashboards, secure authentication, and real-time updates.
- Implemented using Next.js and the MERN stack with role-based access control and email verification.

• Woman Safety Device (Electronic Shop Practice)

2023, RUET

- Designed an ESP32-based safety device integrated with a custom Android SOS app via Bluetooth.
- Implemented live GPS location sharing, incident logging for analysis, and a built-in taser for primary self-defense.

• Nutrition Website (Software Project I)

2023, RUET

- Developed a health-focused web platform for BMI, BMR, and daily calorie estimation based on user activity.
- Implemented OTP-based email verification for secure user registration and login.

EXPERIENCE

• Industrial Trainee

BJIT Ltd.

March 2025

Dhaka, Bangladesh

- Completed industrial training at BJIT Academy with hands-on exposure to SDLC and Agile/Scrum practices.
- Developed Python-based applications including an OOP-driven Library Management System and a Task Manager with JSON persistence.
- Gained practical experience with Git/GitHub, software quality assurance concepts, and team-based software development.

SKILLS

- **Programming:** Python, MATLAB, C++, JavaScript
- **Signal Processing:** MNE-Python, EEG preprocessing, spectral analysis
- **Research Tools:** Git, GitHub, L^AT_EX, Overleaf, Jupyter, VS Code
- **Engineering & Design Tools:** Proteus, EasyEDA, AutoCAD, AutoCAD Electrical
- **Web Development:** MERN Stack, Next.js, HTML, CSS
- **Documentation & Productivity Tools:** L^AT_EX, Microsoft Office

ACHIEVEMENTS

- Merit-Based Scholarship during BSc. from the Directorate of Technical Education, Bangladesh.
- General Scholarship (SSC) and Talentpool Scholarship (JSC & PSC).
- Second Runner-Up (Junior Category) in the Regional Bangladesh Mathematical Olympiad (BDMO) and participant in the National Round.
- Champion, Inter-Department Volleyball Tournament, RUET.

LEADERSHIP, SERVICE & ACTIVITIES

• Class Representative (2nd Year)

Department of Electrical and Computer Engineering

- Represented the class in academic and administrative matters and coordinated communication between students and faculty.

• Volunteer

trECEnto, American Center Pop-Up & Inter-Department Project Showcasing

- Supported event organization, coordination, and logistics for departmental programs, the U.S. Embassy-supported American Center Pop-Up, and university project showcases.

• Sports

Department of Electrical and Computer Engineering

- Represented the department in inter-department football, volleyball, fresher's football, and cricket tournaments.

REFERENCES

Fariya Tabassum

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Rana Mozumder

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